

CLINICAL STUDY · BERTONE ET AL. · 138 PERFORMANCE HORSES

Built on evidence. Designed for scrutiny.

The study design rigor behind equicenta® CTM (eCTM)

138performance horses
enrolled**8**U.S. specialty equine
practices**100%**completed 12-week
primary outcome**1.9:1**eCTM : standard of care
allocation achieved

Why this study stands up to review

Prospective, multicenter, controlled

Real-world pragmatic design across 8 specialty performance-horse practices with a flexible clinician-selected standard-of-care comparator arm — reflecting how biologics are actually used in equine sports medicine.

Independent oversight

IACUC approved (ETCR-2024-0329) and registered in the AVMA Veterinary Clinical Trials Registry (VCT #26005934). Multicenter director and third-party adjudicator were independent of the sponsor.

Blinded, third-party biostatistics

Analyses performed by an independent biostatistician using validated mixed-effects models (R: lme4, lmerTest, emmeans). Data analysts and kinematic reviewers were blinded to treatment assignment.

CONSORT-guided reporting

Trial conducted and reported under CONSORT 2025 guidelines with a pre-specified superiority framework, quasi-randomized allocation within practice, and prospective outcome definitions.

Standardized, validated outcomes

AAEP lameness grade plus validated 0–4 pain and swelling scores. Project veterinarians completed consensus training on live cases to minimize inter-observer variability.

Objective secondary endpoints

Blinded quantitative ultrasound (Tissue Remodeling Score) and sensor-based kinematic gait analysis (Sleip.com) provided assessor-independent confirmation of the clinical findings.

Six safeguards against bias

SELECTION BIAS

Pre-defined eligibility, consecutive enrollment, stratified allocation

PERFORMANCE BIAS

Same veterinarian performed all follow-ups for a given case

DETECTION BIAS

Blinded ultrasound analysis, kinematic review, and data entry

FINANCIAL BIAS

No outcome-contingent compensation; signed COI declarations

EXPECTATION BIAS

Standardized outcome definitions with consensus scoring training

ATTRITION BIAS

100% follow-up at 12 wk, 98% at 26 wk; ITT and per-protocol

A single, off-the-shelf injection of eCTM — tested under real-world conditions, held to CONSORT standards.

Findings supported by primary clinical outcomes, blinded imaging, and objective kinematic gait analysis.

Equine Performance Labs